Smart Study Assistant Agent 📚

AI-Powered Personalized Learning System

Developer : Ketan Dutt

Program : Google GenAI Academy 2026

Transforming Static Materials into Interactive Learning

Upload Materials

PDFs or notes

Ask Questions

AI explanations

Generate Quizzes

Exam simulations

Track Weak Areas

Automatic detection

Revision Plans

Personalized schedules

Acts as a **personal AI tutor**, adapting to each student's learning needs.

Students Face Critical Challenges

Current Problems

- Complex topics difficult to understand
- No personalized guidance
- Inefficient revision strategies
- Weak areas go untracked

Traditional Methods

- Static ❌
- Non-adaptive ❌
- Time-consuming ❌

Our AI-Powered Solution

Interactive Learning

Converts notes into dynamic experience

Simple Explanations

Complex concepts made easy

Active Recall

Quizzes for retention

Smart Tracking

Weak topics identified

Structured Plans

Revision schedules created

Result: **Smart, adaptive, and efficient learning**

How It Meets Build Criteria

Google Gemini API

Generative AI implementation

AI Agent Workflow

Complete pipeline architecture

Real-World Data

PDF processing capability

Actionable Outputs

Answers, quizzes, insights, plans

End-to-End Application

UI + backend integration

Unique Value Proposition

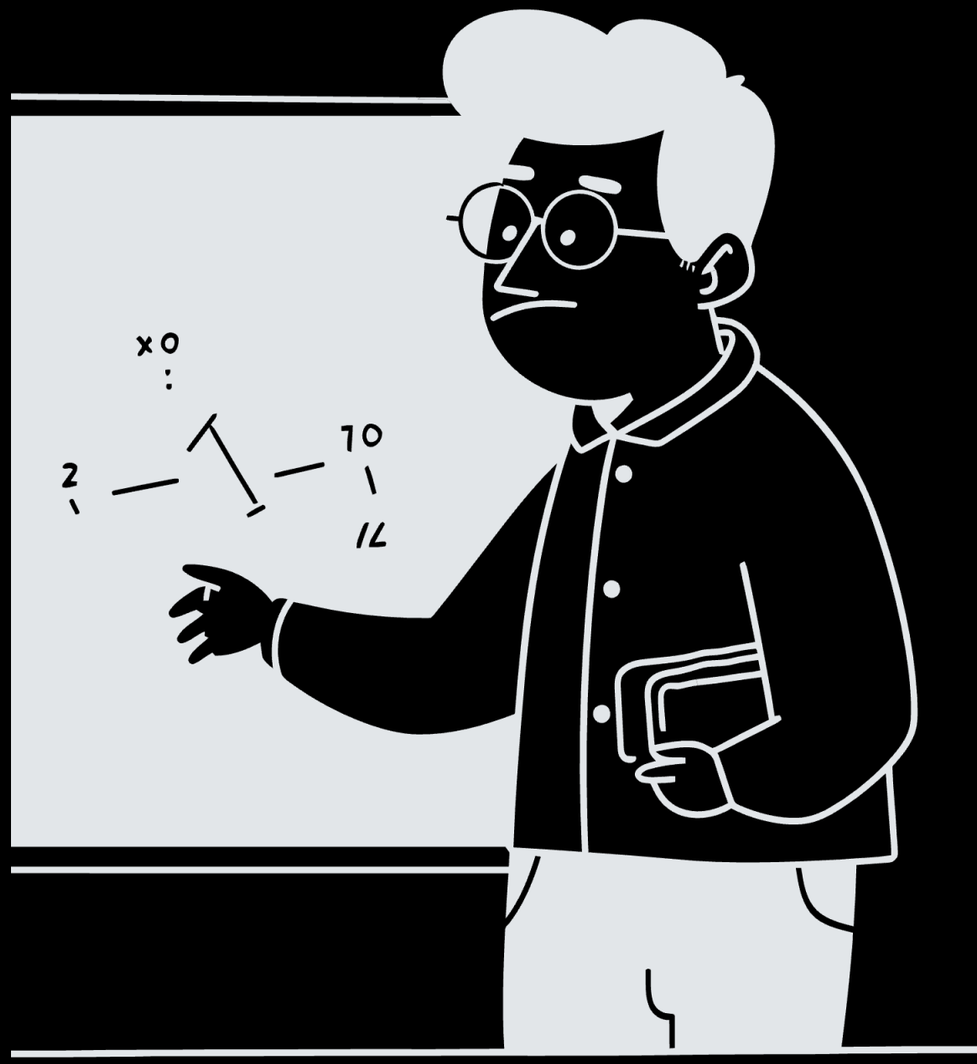
Complete Ecosystem
Not just a chatbot—full learning platform

Personalized Tutor
Adapts to individual needs

Weak Area Detection
Automatic topic identification

Smart Revision
Intelligent planning system

Special Feature: "Explain Like I'm 10" Mode
Makes learning accessible to beginners



Comprehensive Feature Set

PDF Upload
Text extraction

AI Q&A
Smart answering

Quiz Generator
MCQs + explanations

Exam Mode
Test simulation

Weak Tracking
Topic detection

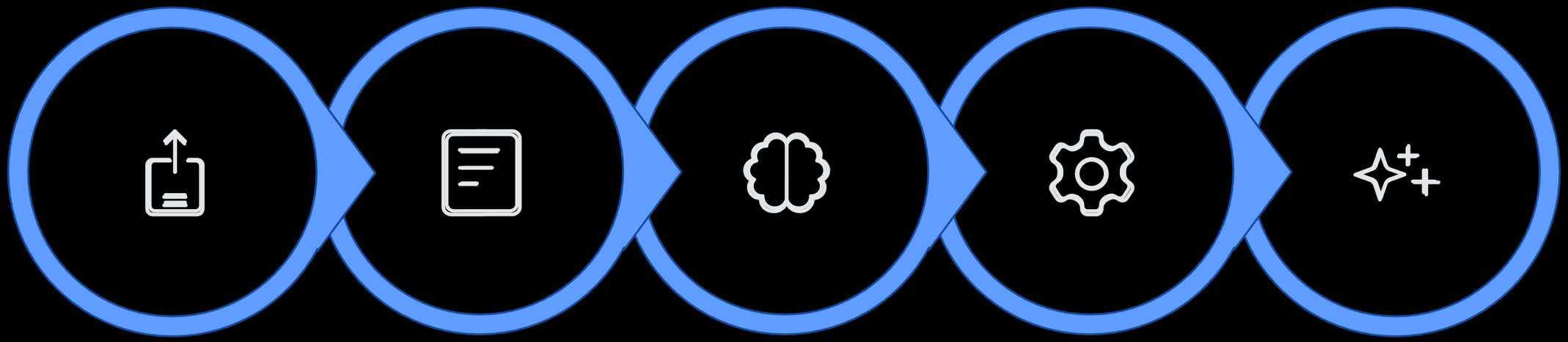
Revision Planner
Smart scheduling

Flashcards
Interactive system

Summary Generator
Concise notes

Progress Dashboard
Visual tracking

System Workflow



Upload PDF | Extract Text | Analyze (Gemini) | Select Feature | Generate Output

Complete **AI Agent Pipeline**

Technology Stack

Core Technologies

- Google Gemini API (GenAI)
- Python backend
- Streamlit UI framework
- PyPDF for PDF parsing
- python-dotenv for security
- Pandas for analytics
- Google Cloud Run deployment

AI Architecture

- Context selection (chunking + ranking)
- Prompt engineering
- Gemini API integration
- JSON parsing & validation
- Retry logic for robustness
- Caching for performance
- Error handling

Performance Dashboard & Impact

80%

Score Tracking

Real-time percentage display

3-5

Day Plans

Automated study schedules

100%

Focus Areas

Weak topics prioritized

Progress Insights

Score tracking, weak topics, mistake frequency, visual progress

Smart Revision

3–5-day plans, daily goals, actionable tasks, topic prioritization

Flashcards System

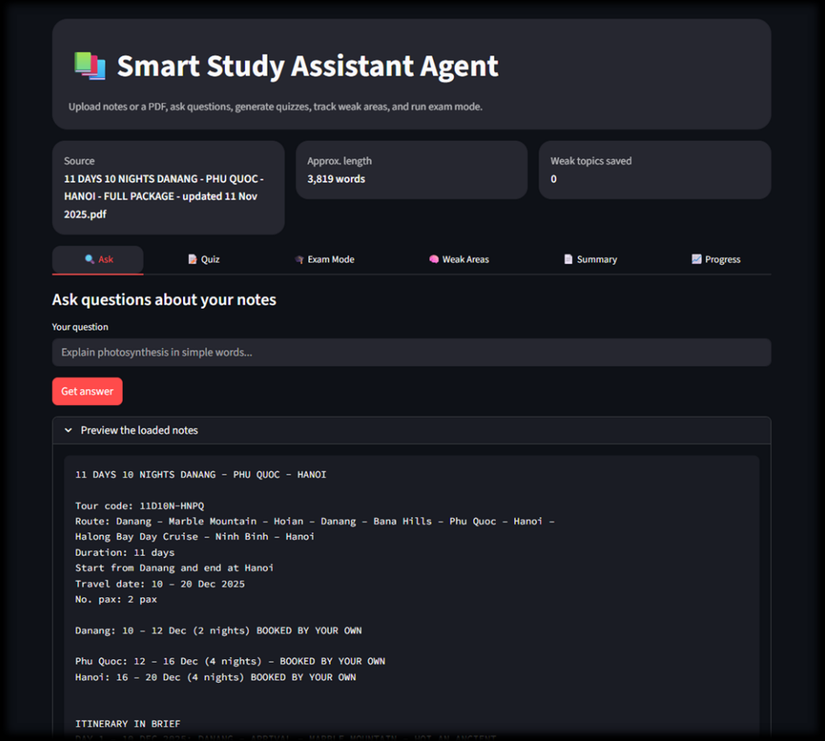
AI-generated cards, weak topic focus, interactive reveal, memory retention

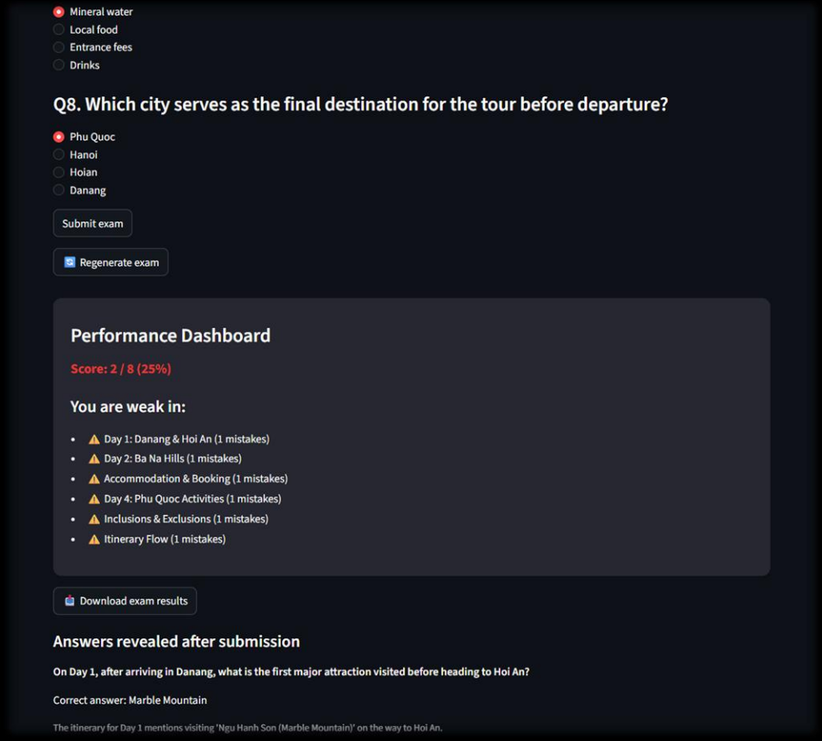
Data-Driven Learning

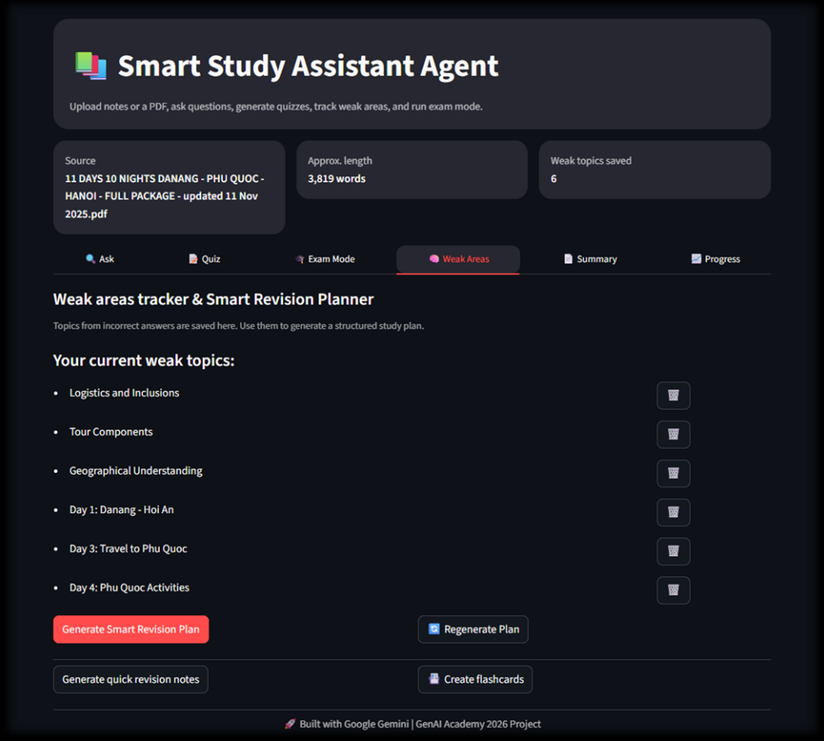
Quiz scores, performance trends, improvement tracking

Intuitive Design in Action

Explore the seamless experience of our Smart Study Assistant, designed for ease of use.







Our user interface brings advanced AI features to your fingertips, simplifying complex study tasks and enhancing your learning journey.

🙏 Thank You

Thank you for your time and consideration.

📚 Smart Study Assistant Agent

Generative AI transforms traditional learning into personalized education, improving student outcomes with scalable AI-driven systems.

Live Demo | GitHub Repository

🔗 Connect with me

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🚀 Built using Google Gemini | GenAI Academy 2026